



ProDenCeR:

a python code to project first-principles
densities onto harmonics and representations

Luca Buiarelli, Hyeonseo Park, Seongjoo Jung, Turan Birol

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March 20



Chemical Engineering
& Materials Science

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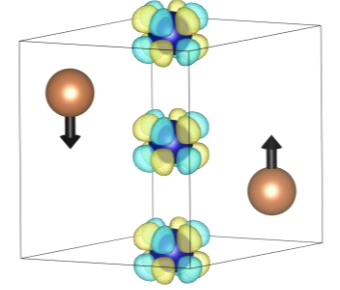
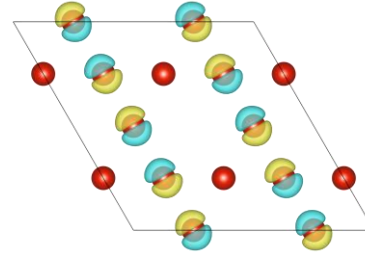
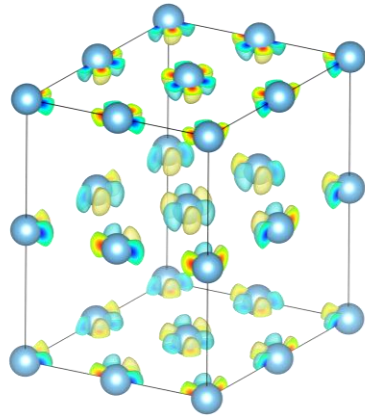
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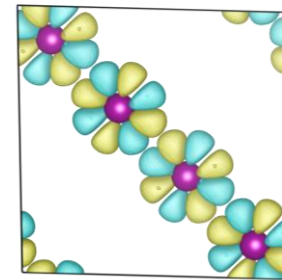
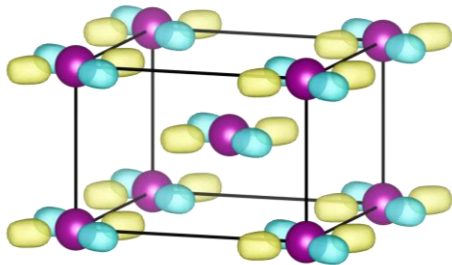


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Motivation



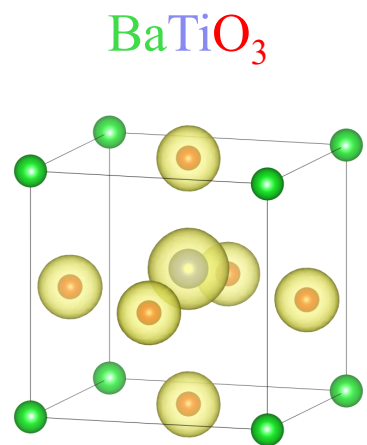
- **Symmetry breaking** not always can be represented as lattice/atomic distortions
- Order parameters can be 'hidden' in the **charge and spin densities**
- Can be connected to **macroscopic observables** through symmetry arguments



ProDenCeR: the two main functions



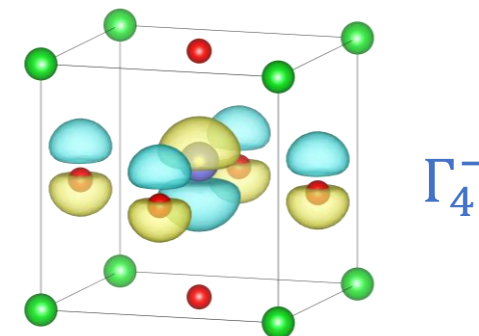
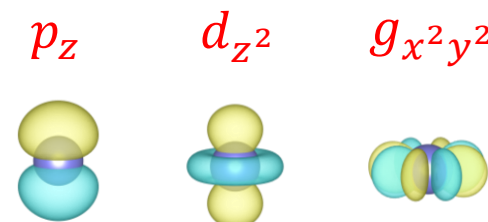
Project Densities onto Cubic/tesseral harmonics & space group Representations



Charge/Spin
Density

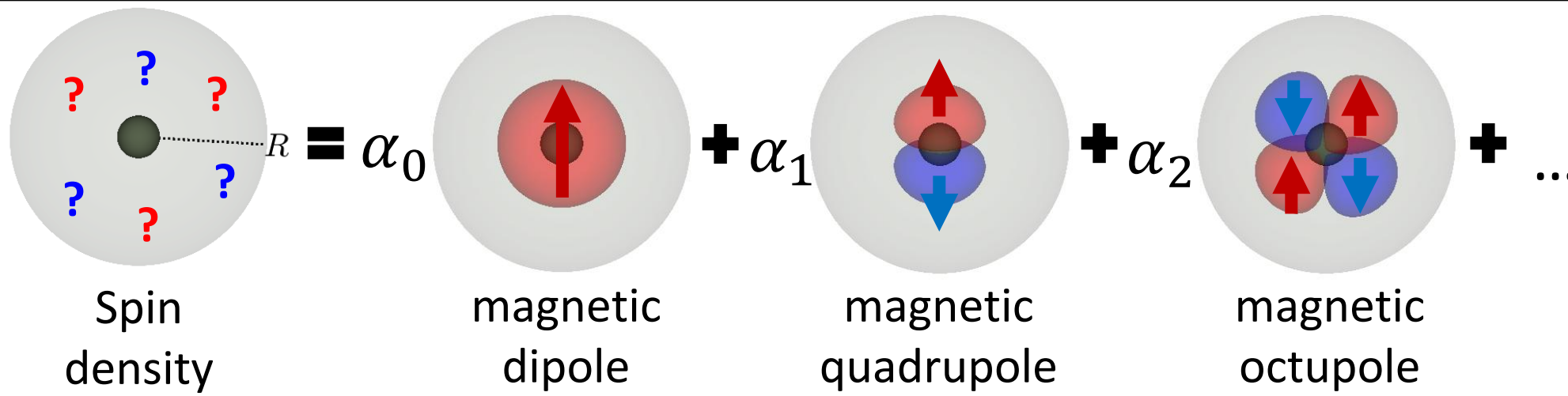
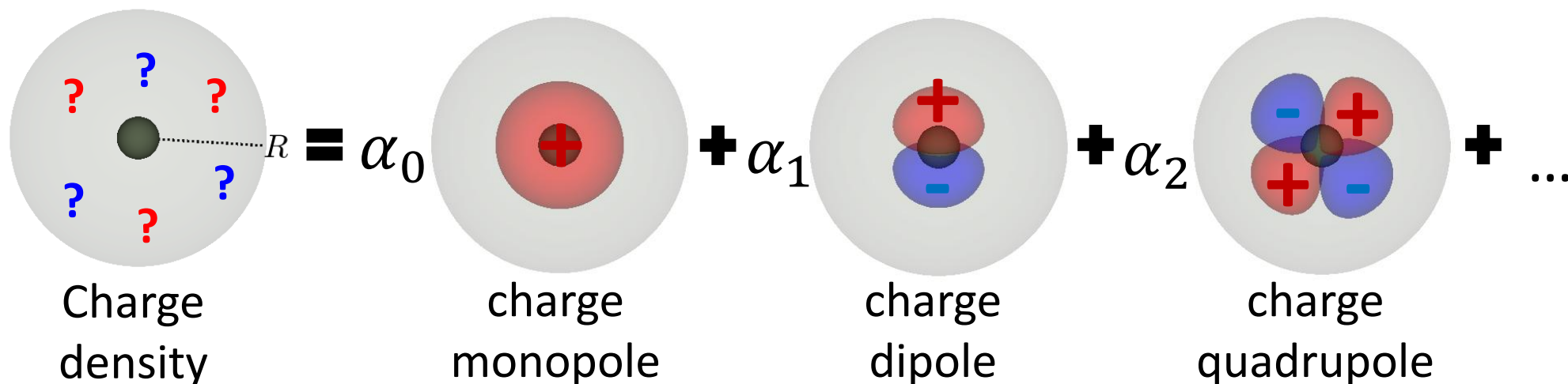
Atomic multipoles decomposition:
`project_harmonics()`

Space group irreps decomposition:
`project_irreps()`



Atomic multipoles: cubic/tesseral harmonics

$$\alpha_{\ell m} = \sqrt{\frac{4\pi}{2\ell + 1}} \int_{|\mathbf{r}-\mathbf{r}_0| < R} \underbrace{d^3r \rho(\mathbf{r})}_{\text{Density}} \underbrace{T_{\ell m}(\mathbf{r}) r^\ell}_{\text{Cubic harmonics}}$$



Projecting onto the irreps

Projection formula:

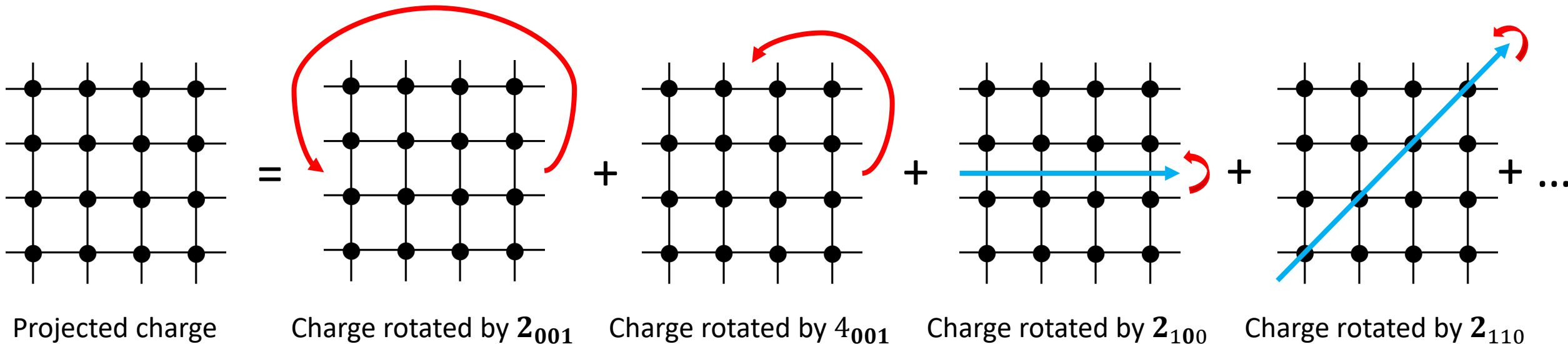
$$\rho^\alpha(\mathbf{r}) = \frac{1}{|G|} \sum_{g \in G} \underbrace{\chi^\alpha(g)}_{\text{Character}} \underbrace{\rho(g\mathbf{r})}_{\text{Rotated charge}}$$

Irrep label

Symmetry elements

Character table for 4/mmm
(only inversion-even irreps)

χ	1	2_{001}	4_{001}	2_{100}	2_{110}
Γ_1^+	1	1	1	1	1
Γ_2^+	1	1	-1	1	-1
Γ_3^+	1	1	1	-1	-1
Γ_4^+	1	1	-1	-1	1
Γ_5^+	2	-2	0	0	0



Projecting onto the irreps

Projection formula:

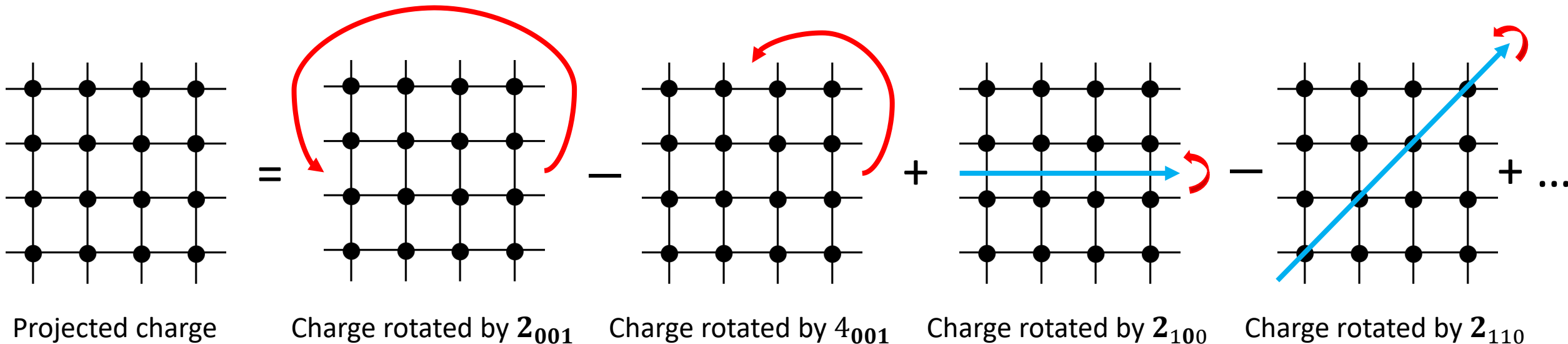
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Irrep label

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Character table for 4/mmm
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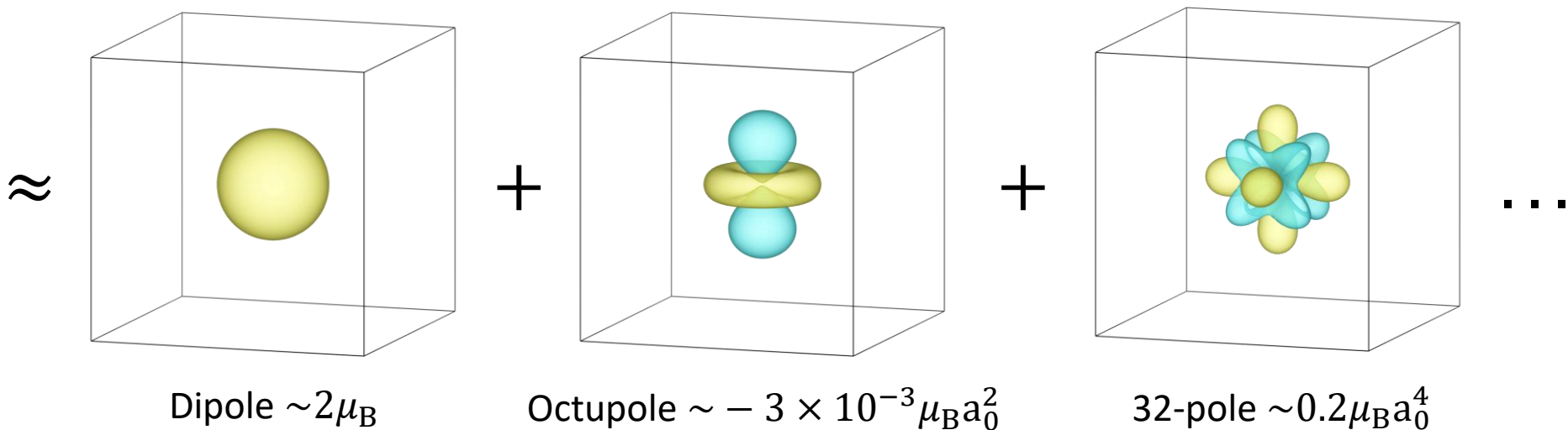
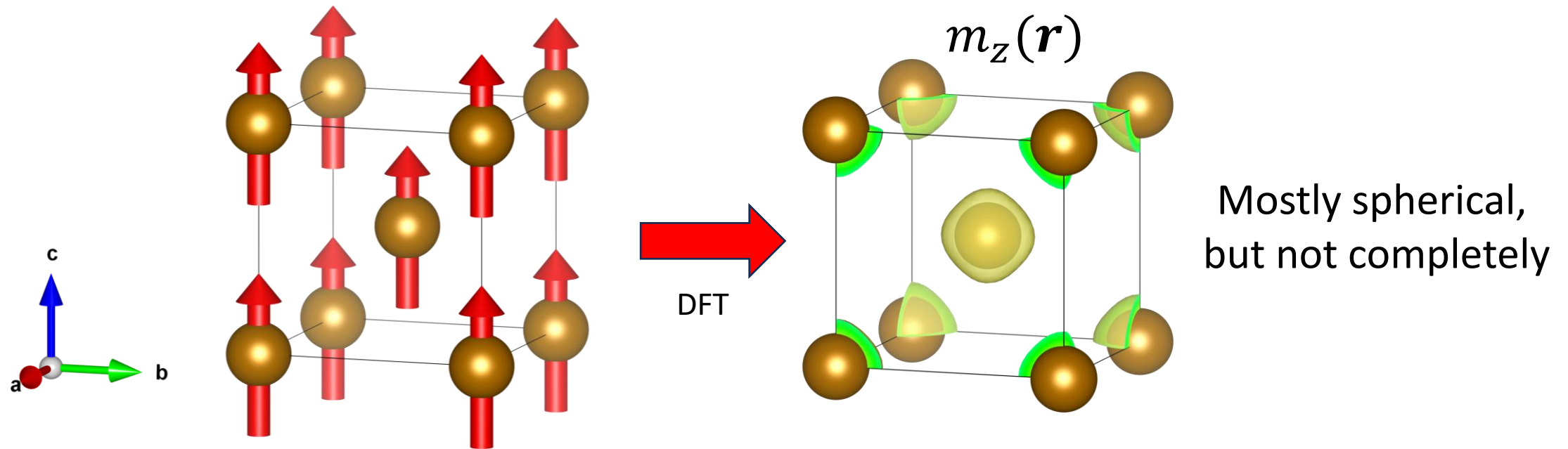
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Γ_4^+	1	1	-1	-1	1
Γ_5^+	2	-2	0	0	0



Examples

Atomic multipoles: unconventional magnetism

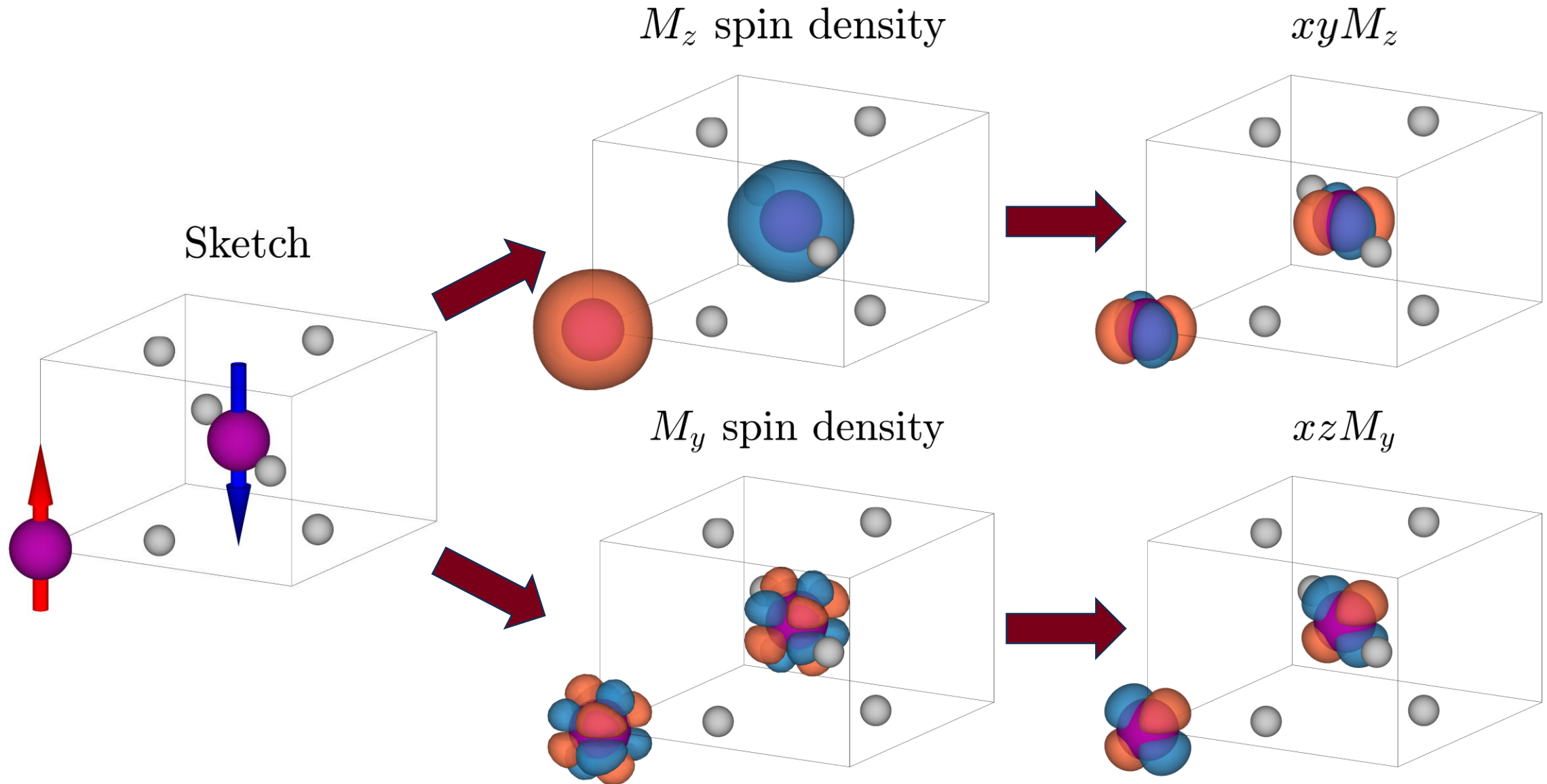
Ferromagnetic Iron



Altermagnetism in MnF_2

S. Bhowal and N. Spaldin, Phys. Rev. X **14**, 011019 (2024)

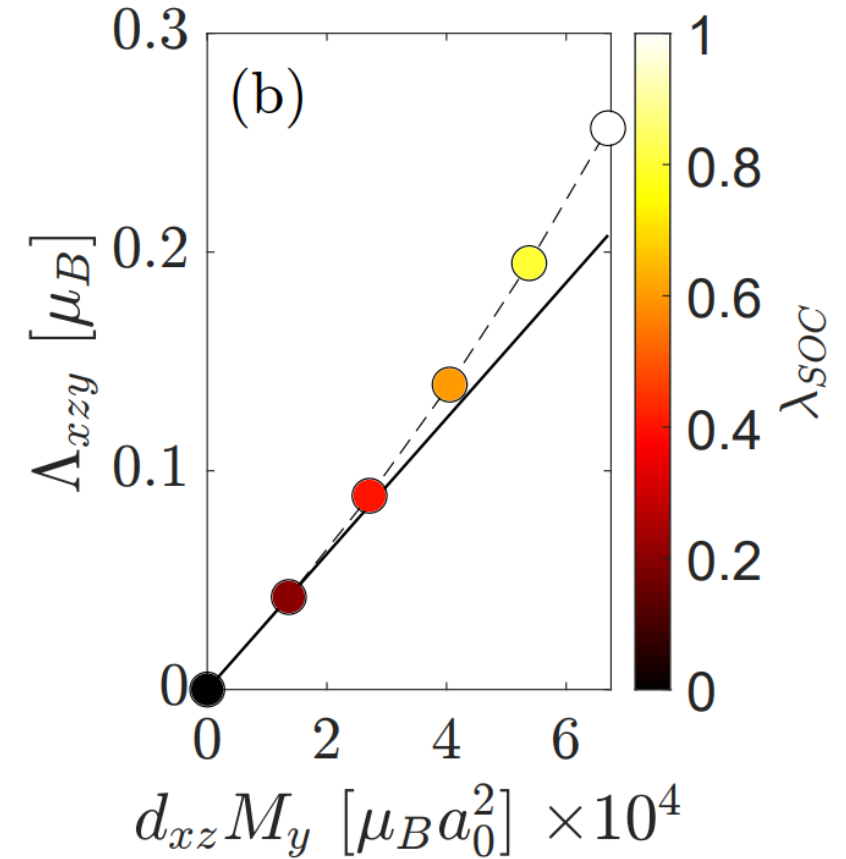
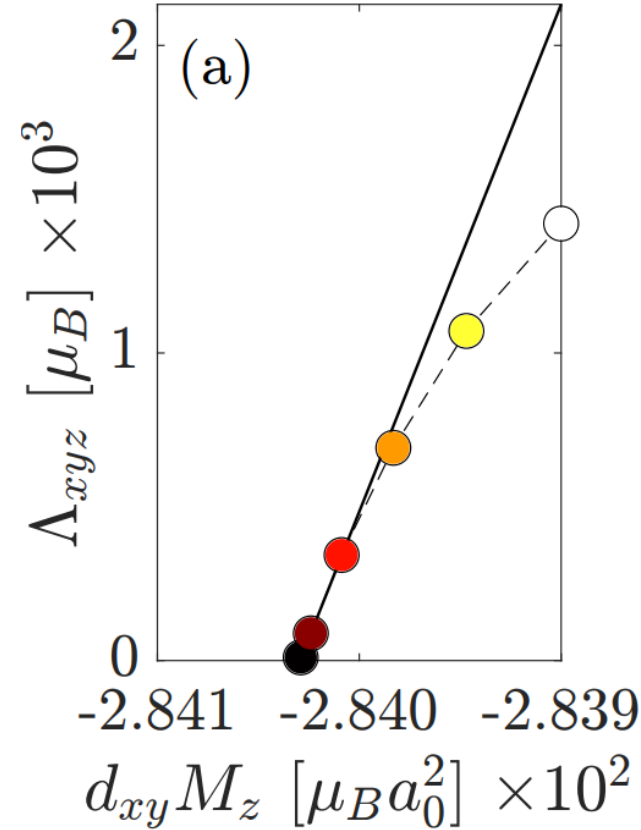
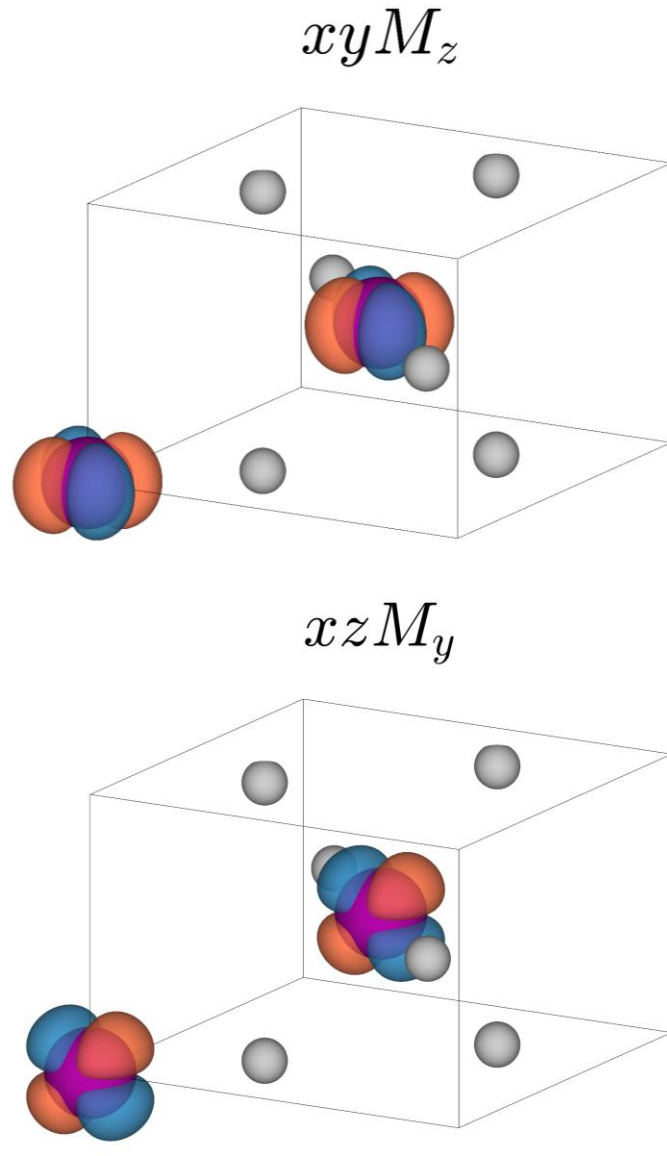
LB, R.M. Fernandes, T. Birol, Phys. Rev. B **112**, 224442 (2025)



Altermagnetism in MnF_2

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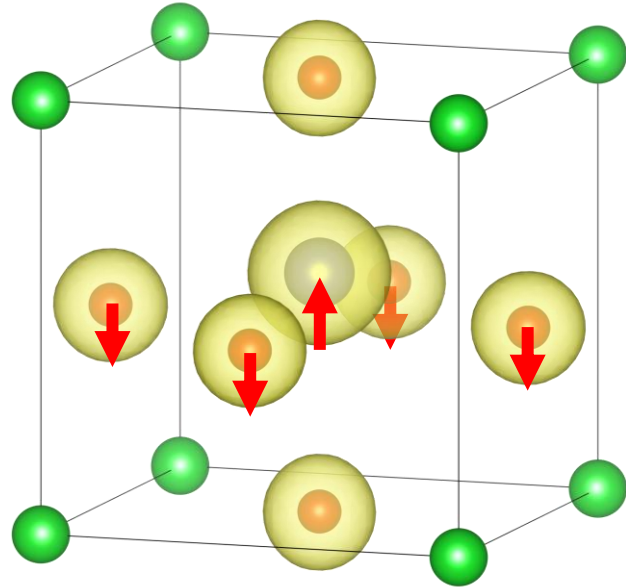
Correlated with macroscopic
observables such as **piezomagnetism**

Examples

Space group representations: ferroelectricity and chirality

Ferroelectric BaTiO₃

Cubic $m\bar{3}m$ → Tetragonal $4mm$

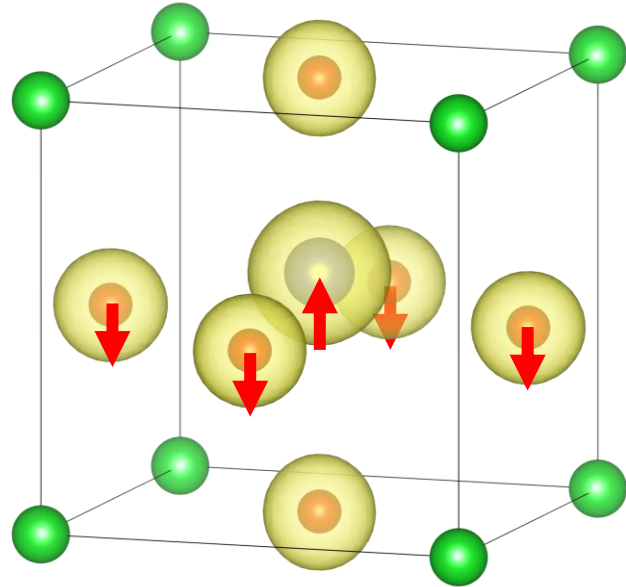


Partial character table for $m\bar{3}m$

χ	$\bar{1}$	3_{111}	4_{001}	m_{001}	m_{110}
Γ_1^+	1	1	1	1	1
Γ_3^+	2	-1	0	2	0
Γ_4^-	-3	0	1	1	1

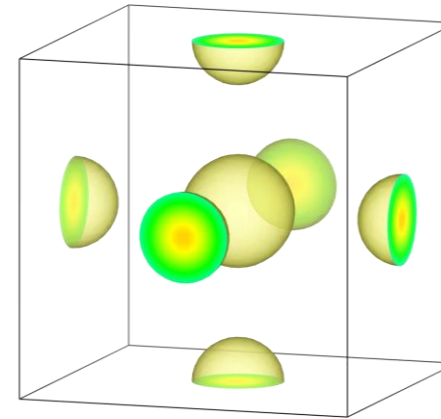
Ferroelectric BaTiO₃

Cubic $m\bar{3}m$ → Tetragonal $4mm$



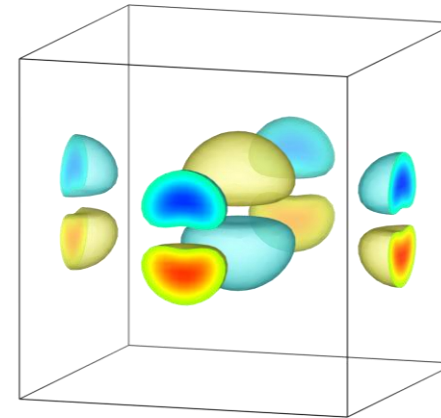
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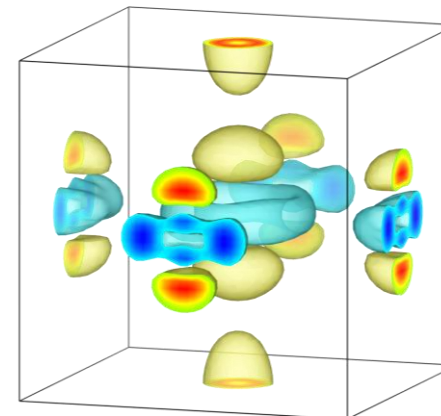
Γ_1^+

trivial...



Γ_4^-

primary OP



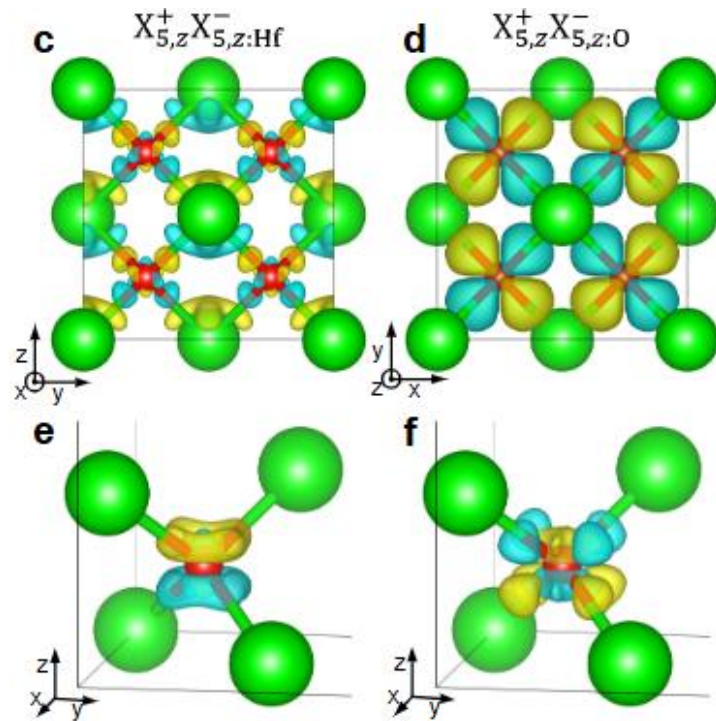
Γ_3^+

secondary OP

Inversion breaking as the SOP

Ferroelectricity in HfO_2 caused by the condensation of several **non-polar** modes

Projection onto Γ_4^-

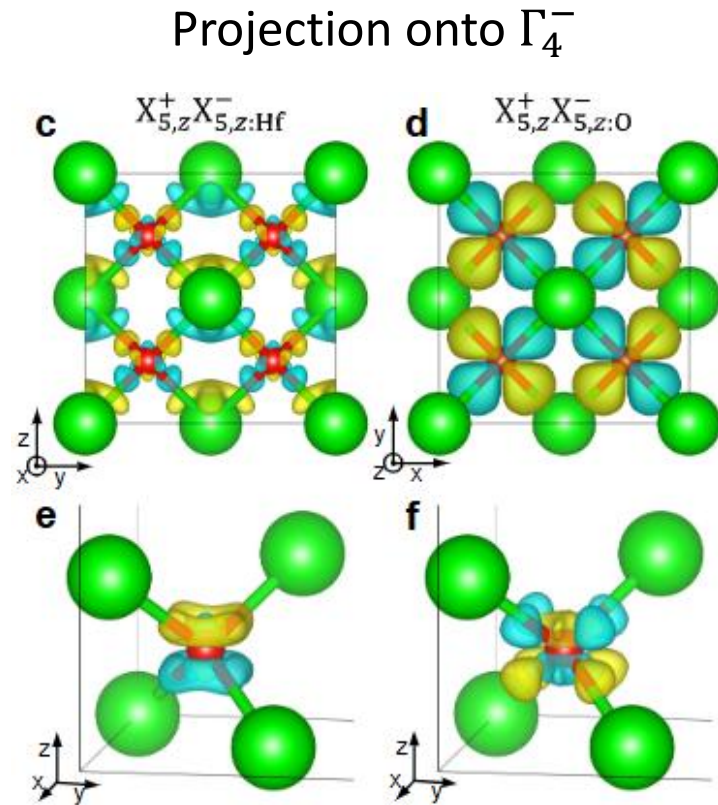


S. Jung, T. Birol, Adv. Mat. (2026)

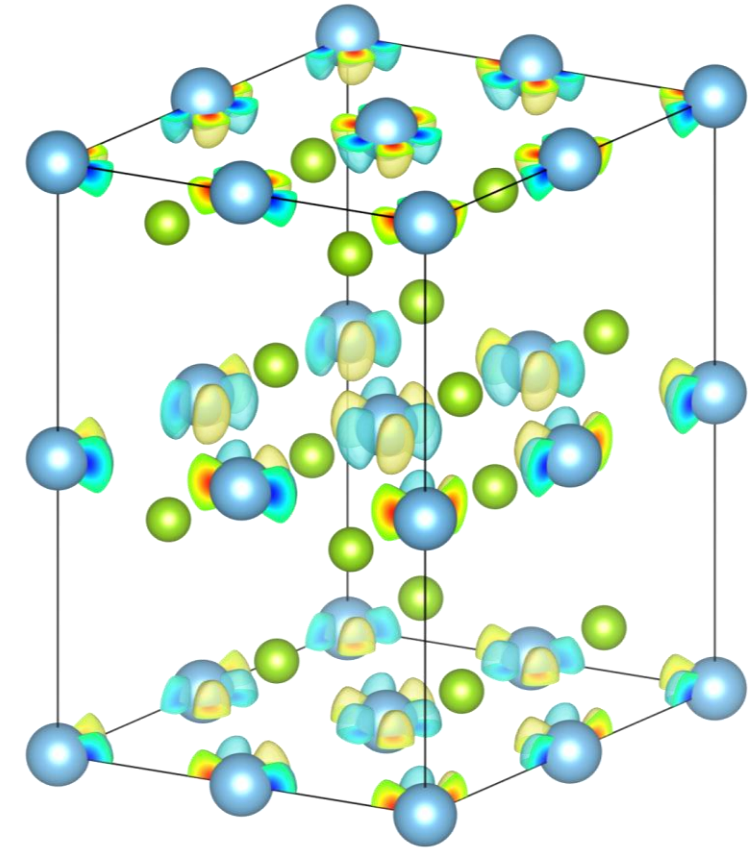
S. Jung, T. Birol, arXiv: 2512.00628 (2025)

Inversion breaking as the SOP

Ferroelectricity in HfO_2 caused by the condensation of several **non-polar** modes



Projection onto Γ_1^-



Chiral CDW phase in TiSe_2
Each order parameter by itself does **not** break inversion

S. Jung, T. Birol, Adv. Mat. (2026)

S. Jung, T. Birol, arXiv: 2512.00628 (2025)

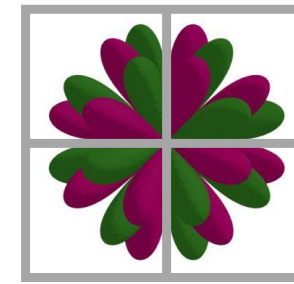
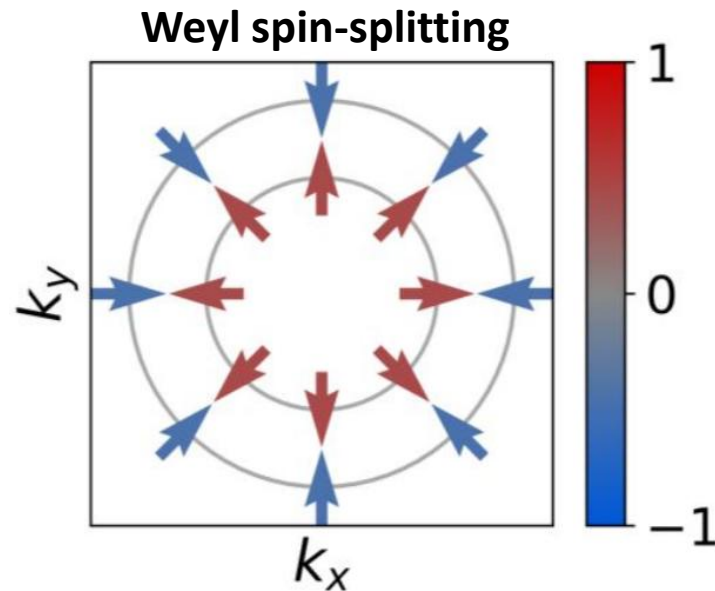
H. Park, S. Jung and T. Birol (in preparation)

Electric toroidal monopole

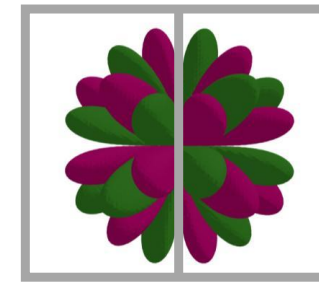
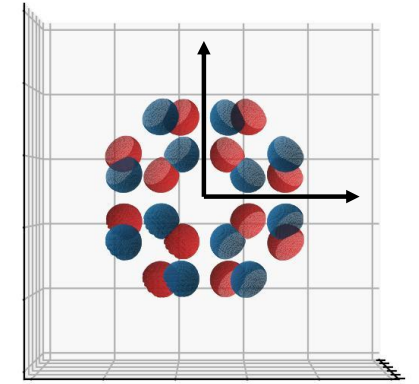
What if Γ_1^- was the primary OP?

The electric toroidal monopole can be represented as the electric 512-pole.

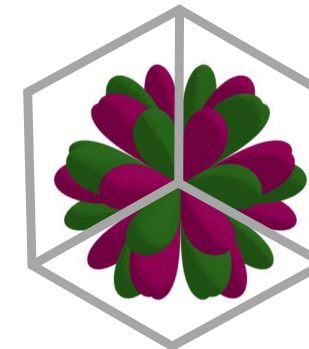
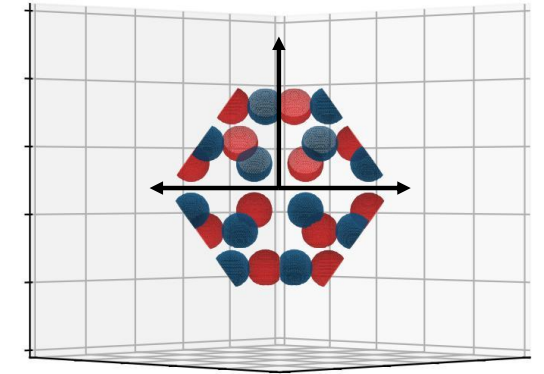
The charge of LaCeBe_{26} has a non-zero projection onto the Γ_1^- irrep.



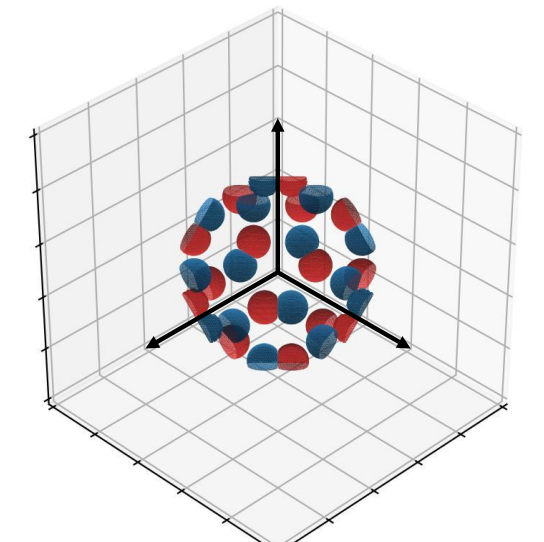
[100]



[110]



[111]



Summary



- Developed open-source Python code to do several operations with **real-space charge/spin densities** obtained from DFT
- **Atomic multipole** projection
- **Space group irrep** projection
- **Purpose:** extract “hidden” order parameters and connect to observables

Github:



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Appendix

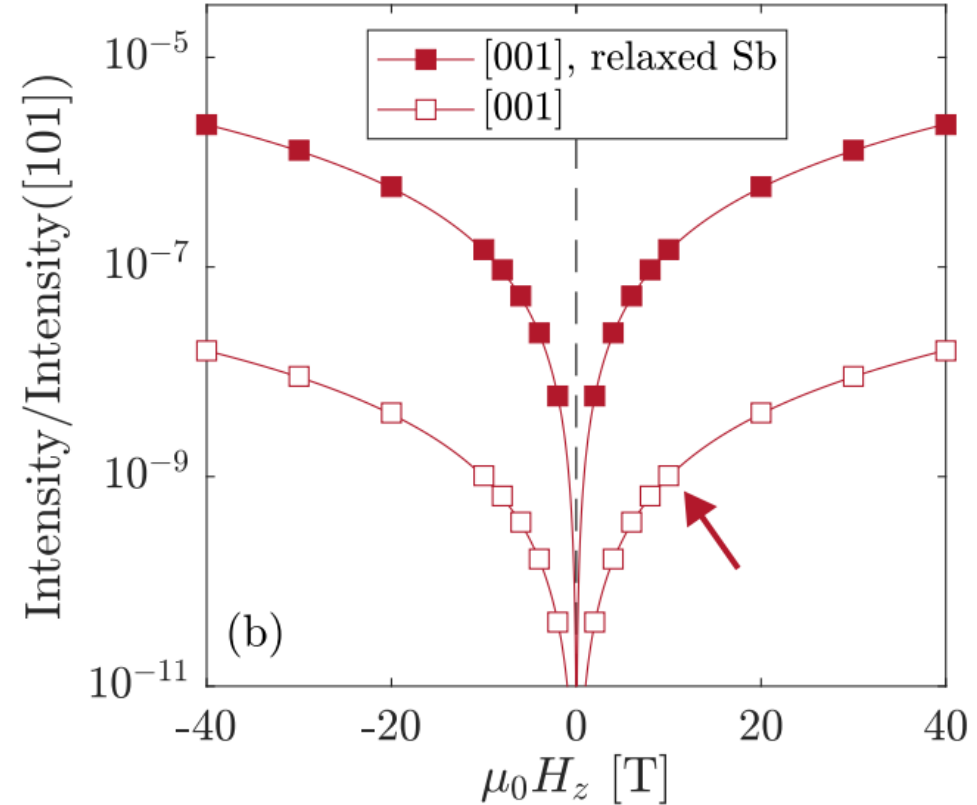
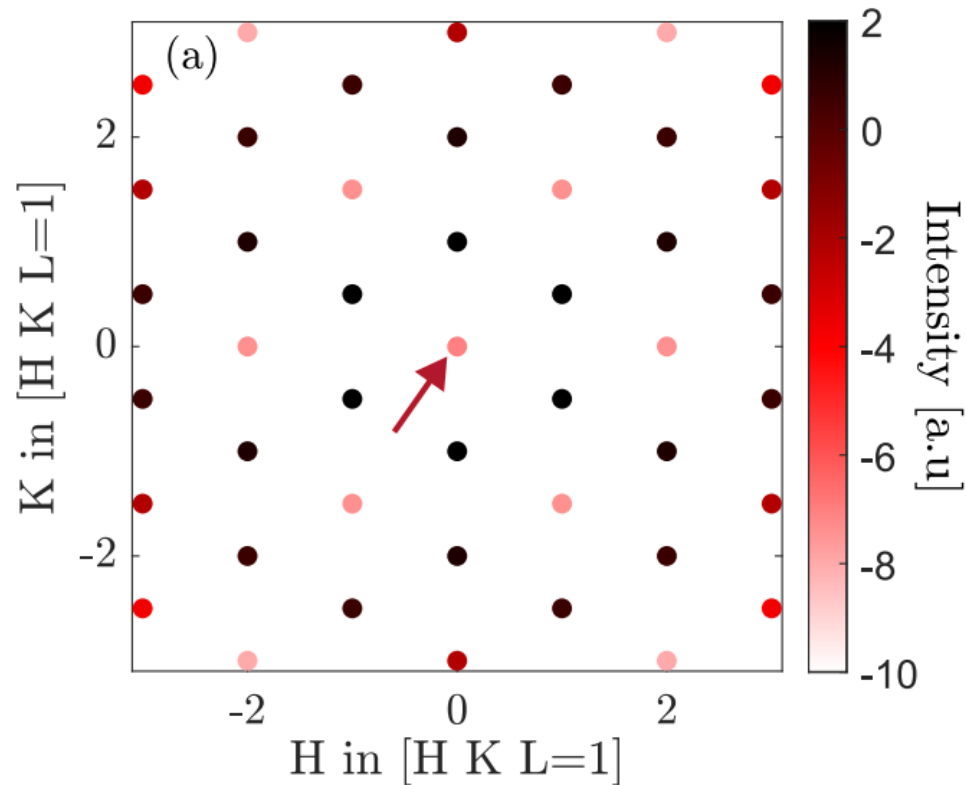
Details on theory and implementation

Simulating XRD patterns

LB, R.M. Fernandes, T. Birol (in preparation)

Fourier transform the charge density onto reciprocal lattice to predict XRD patterns

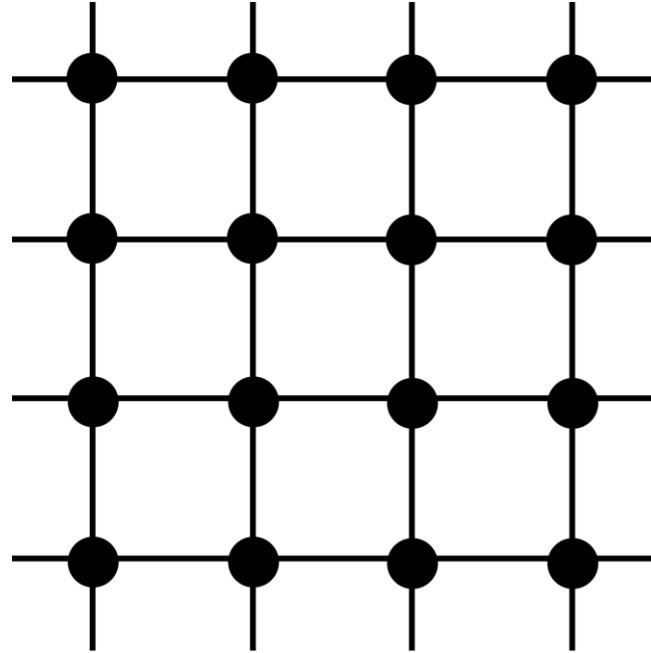
Example: applied magnetic field to g-wave altermagnet breaks extinction rules



Importing the DFT densities



DFT calculation



Density defined on a
 $N_x \times N_y \times N_z$
grid

Charge density

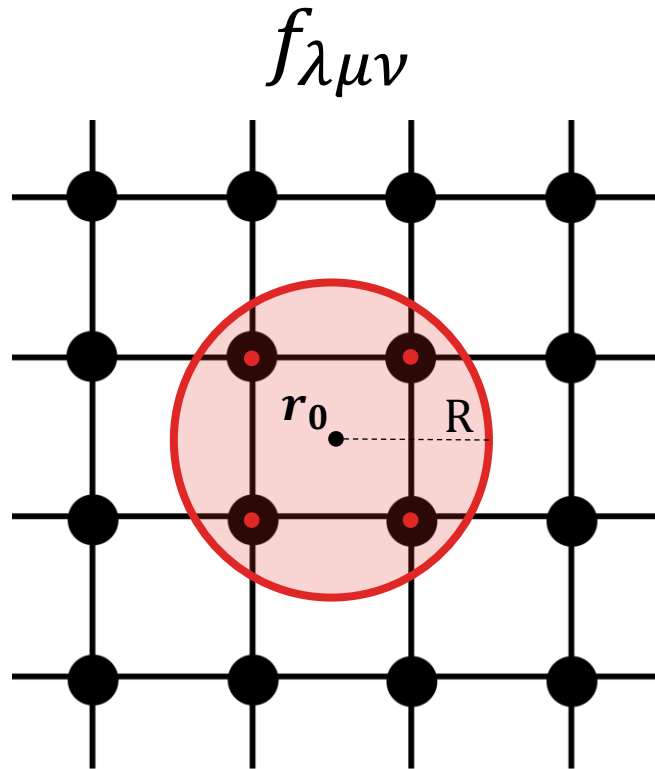
$$\rho_{\lambda\mu\nu}$$

Spin density

$$m_{\lambda\mu\nu}^i$$

Numerical integration

Integrate a function f inside a sphere centered at $\mathbf{r}_0$ with radius R .

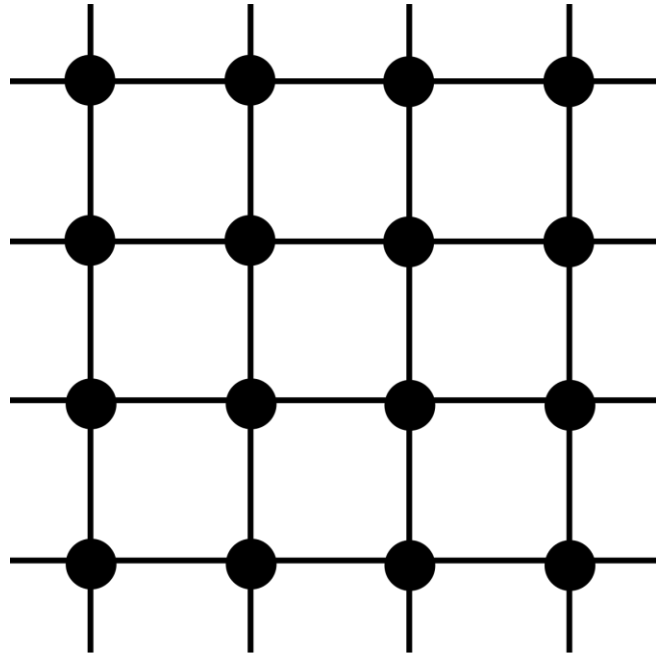


- Use the real-space grid to define a “mask” of points that are inside the sphere
- Integrate the “masked” function

$$I[f; R, \mathbf{r}_0] = \int_{|\mathbf{r}-\mathbf{r}_0| < R} d^3r f(\mathbf{r})$$
$$\approx \sum_{\lambda\mu\nu} f(\mathbf{r}_{\lambda\mu\nu} - \mathbf{r}_0) dV \quad \text{for } |\mathbf{r} - \mathbf{r}_0| < R$$

$$dV = V/N_x N_y N_z$$

Creating a real-space grid



We create a real-space grid of the same size
 $N_x \times N_y \times N_z$

First, we create “reduced grids”

$$\mathbf{r}_{\lambda\mu\nu}^{\text{red}} = \left(x_{\lambda\mu\nu}^{\text{red}}, y_{\lambda\mu\nu}^{\text{red}}, z_{\lambda\mu\nu}^{\text{red}} \right)^T$$
$$= \left(\frac{\lambda}{N_x}, \frac{\mu}{N_y}, \frac{\nu}{N_z} \right)^T \quad \text{for} \quad \begin{cases} \lambda = 0, \dots, N_x - 1, \\ \mu = 0, \dots, N_y - 1, \\ \nu = 0, \dots, N_z - 1. \end{cases}$$

Then, we scale them with the lattice

$$\mathbf{r}_{\lambda\mu\nu} = L \cdot \mathbf{r}_{\lambda\mu\nu}^{\text{red}} = \left(x_{\lambda\mu\nu}, y_{\lambda\mu\nu}, z_{\lambda\mu\nu} \right)^T$$

Numerical integration of multipoles

Example: charge dipole

$$\sum_{\lambda\mu\nu} \left[\rho_{\lambda\mu\nu} \times x_{\lambda\mu\nu} \right] = \text{Atomic electric dipole at } \mathbf{r}_0$$

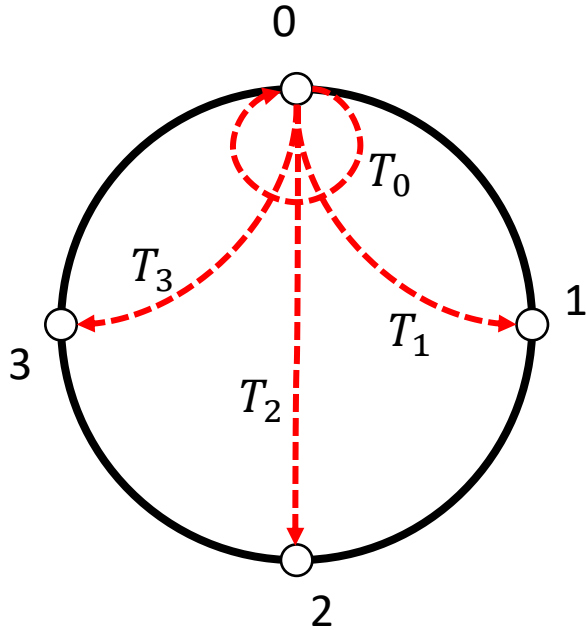
The diagram illustrates the numerical integration of multipoles for a charge dipole. It shows two 4x4 grids of points. The left grid, labeled $\rho_{\lambda\mu\nu}$, has a central 2x2 area of red dots enclosed in a red circle of radius R , with a center point $\mathbf{r}_0$. The right grid, labeled $x_{\lambda\mu\nu}$, has the same central 2x2 area of red dots, but each of the four surrounding black dots is labeled with the number 1, 2, 3, or 4. A large curly bracket on the left groups the sum over $\lambda\mu\nu$, and another large curly bracket on the right groups the product. An equals sign follows, leading to the text "Atomic electric dipole at $\mathbf{r}_0$ ".

Groups and representations: translations

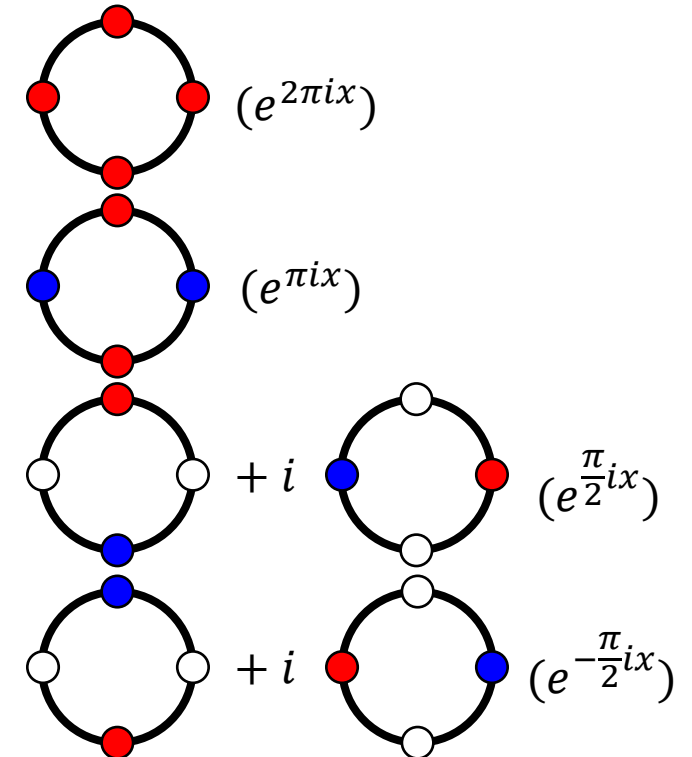
Consider a ring with four points: $x \in \{0,1,2,3\}$.

The translation **operators** are T_0, T_1, T_2, T_3 and form a **group** (identity, inverse, closed).

The character table for the **irreducible representations** (k-points) is



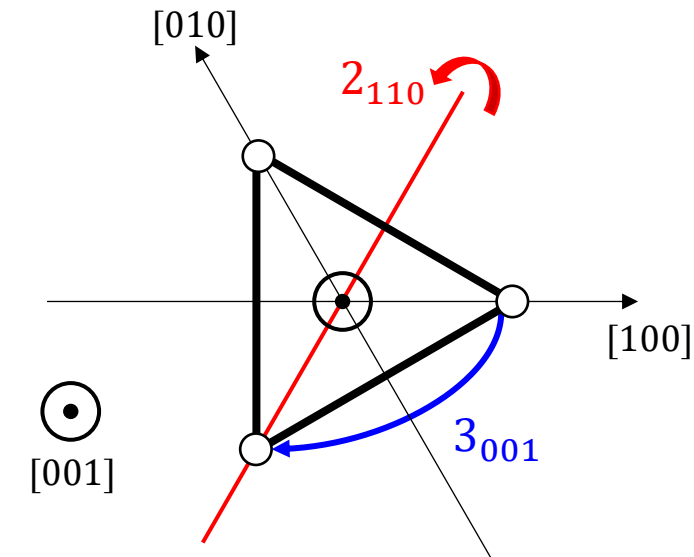
	T_0	T_1	T_2	T_3
k_0	1	1	1	1
k_1	1	-1	1	-1
k_2	1	i	-1	$-i$
k_3	1	$-i$	-1	i



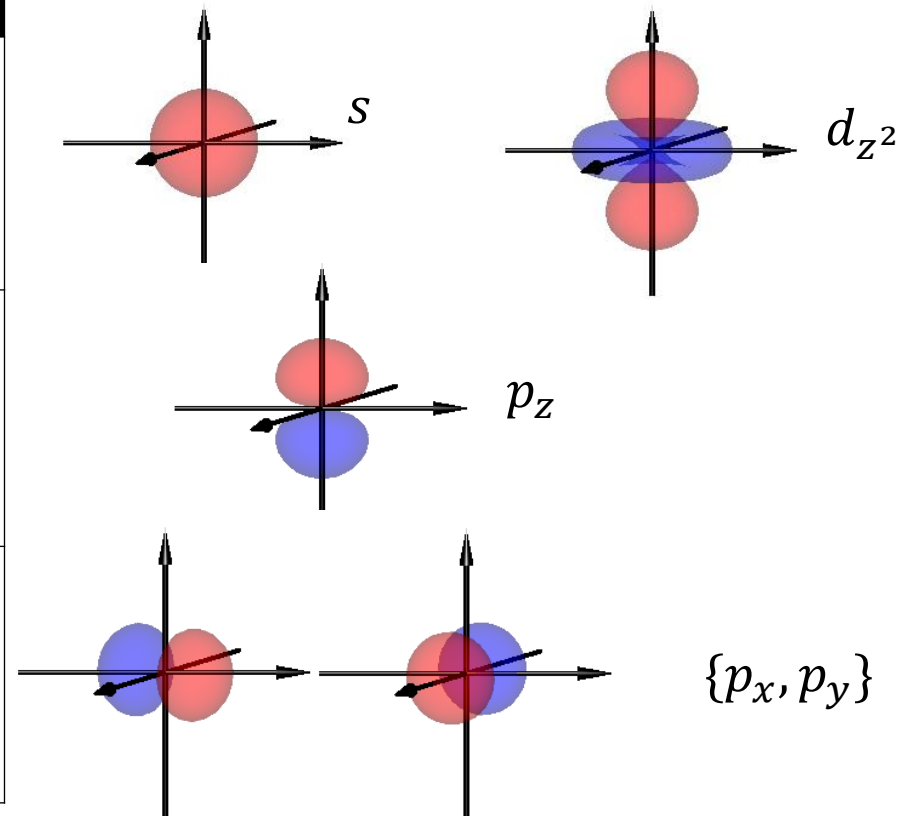
Groups and representations: rotations

Consider the group of symmetries $D_3 = \{1, 3_{001}, -3_{001}, 2_{100}, 2_{110}, 2_{010}\}$

The character table for the **irreducible representations** is

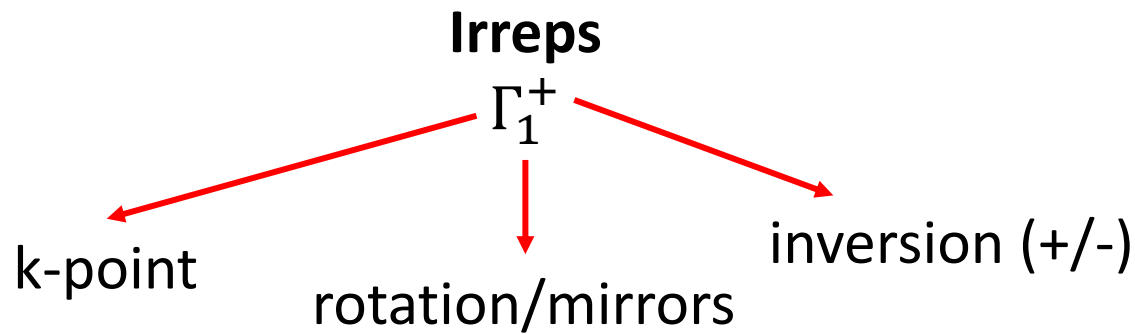


	1	3_{001}	2_{100}
A_1	1	1	1
A_2	1	1	-1
E	2	-2	0



Space Group Irreps

Rotations/mirrors + translations
Space Groups
 (230)



Handle every commensurate k-point!

Thanks to: spglib, spgprep

Table of Space Group Symbols

1	P1	2	P-1	3	P2	4	P2 ₁	5	C2
6	Pm	7	Pc	8	Cm	9	Cc	10	P2/m
11	P2 ₁ /m	12	C2/m	13	P2/c	14	P2 ₁ /c	15	C2/c
16	P222	17	P222 ₁	18	P2 ₁ 2 ₁ 2	19	P2 ₁ 2 ₁ 2 ₁	20	C222 ₁
21	C222	22	F222	23	I222	24	I2 ₁ 2 ₁ 2 ₁	25	Fmm2
26	Fmc2 ₁	27	Pcc2	28	Fma2	29	Fca2 ₁	30	Pnc2
31	Fmn2 ₁	32	Pba2	33	Fna2 ₁	34	Pnn2	35	Cmm2
36	Cmc2 ₁	37	Ccc2	38	Am2	39	Aem2	40	Ama2
41	Aea2	42	Fmm2	43	Fdd2	44	Imm2	45	Iba2
46	Ima2	47	Pmmm	48	Pnnn	49	Pccm	50	Pban
51	Pmma	52	Pnna	53	Pmna	54	Pcca	55	Pbam
56	Pccn	57	Pbcm	58	Pnnm	59	Pmmn	60	Pbcn
61	Pbca	62	Pnma	63	Cmcm	64	Cmce	65	Cmmm
66	Cccm	67	Cmme	68	Ccce	69	Fmmm	70	Fddd
71	Immm	72	Ibam	73	Ibca	74	Imma	75	P4
76	P4 ₁	77	P4 ₂	78	P4 ₃	79	I4	80	I4 ₁
81	P-4	82	I-4	83	P4/m	84	P4 ₂ /m	85	P4/n
86	P4 ₂ /n	87	I4/m	88	I4 ₁ /a	89	P422	90	P42 ₁ 2
91	P4 ₁ 22	92	P4 ₁ 2 ₁ 2	93	P4 ₂ 22	94	P4 ₂ 2 ₁ 2	95	P4 ₃ 22
96	P4 ₃ 2 ₁ 2	97	I422	98	I4 ₁ 22	99	P4mm	100	P4bm
101	P4 ₂ cm	102	P4 ₂ nm	103	P4cc	104	P4nc	105	P4 ₂ mc
106	P4 ₂ bc	107	I4mm	108	I4cm	109	I4 ₁ md	110	I4 ₁ cd
111	P-42m	112	P-42c	113	P-42 ₁ m	114	P-42 ₁ c	115	P-4m2
116	P-4c2	117	P-4b2	118	P-4n2	119	I-4m2	120	I-4c2
121	I-42m	122	I-42d	123	P4/mmm	124	P4/mcc	125	P4/nbm
126	P4/nnc	127	P4/mbm	128	P4/mnc	129	P4/nmm	130	P4/ncc
131	P4 ₂ /mmc	132	P4 ₂ /mcm	133	P4 ₂ /nbc	134	P4 ₂ /hnm	135	P4 ₂ /mbc
136	P4 ₂ /mnm	137	P4 ₂ /nmc	138	P4 ₂ /ncm	139	I4/mmm	140	I4/mcm
141	I4 ₁ /amd	142	I4 ₁ /acd	143	P3	144	P3 ₁	145	P3 ₂
146	R3	147	P-3	148	R-3	149	P312	150	P321
151	P3 ₁ 12	152	P3 ₁ 21	153	P3 ₂ 12	154	P3 ₂ 21	155	R32
156	P3m1	157	P31m	158	P3c1	159	P31c	160	R3m
161	R3c	162	P-31m	163	P-31c	164	P-3m1	165	P-3c1
166	R-3m	167	R-3c	168	P6	169	P6 ₁	170	P6 ₅
171	P6 ₂	172	P6 ₄	173	P6 ₃	174	P-6	175	P6/m
176	P6 ₃ /m	177	P622	178	P6 ₁ 22	179	P6 ₅ 22	180	P6 ₂ 22
181	P6 ₄ 22	182	P6 ₃ 22	183	P6mm	184	P6cc	185	P6 ₃ cm
186	P6 ₃ mc	187	P-6m2	188	P-6c2	189	P-62m	190	P-62c
191	P6/mmm	192	P6/mcc	193	P6 ₃ /mcm	194	P6 ₃ /mmc	195	P23
196	F23	197	I23	198	P2 ₁ 3	199	I2 ₁ 3	200	Fm-3
201	Pn-3	202	Fm-3	203	Fd-3	204	Im-3	205	Pa-3
206	Ia-3	207	P432	208	P4 ₂ 32	209	F432	210	F4 ₁ 32
211	I432	212	P4 ₃ 32	213	P4 ₁ 32	214	I4 ₁ 32	215	P-43m
216	F-43m	217	I-43m	218	P-43n	219	F-43c	220	I-43d
221	Pm-3m	222	Pn-3n	223	Fm-3n	224	Pn-3m	225	Fm-3m
226	Fm-3c	227	Fd-3m	228	Fd-3c	229	Im-3m	230	Ia-3d